9.12 Random Subsets

WORKSHEET

The question

The set S is the set of all integers from 1 to n. The set T is the set of all distinct subsets of S, including the empty set $\varnothing$ and S itself. Show that T contains exactly 2^n sets.

The sets $A_1, A_2, \ldots, A_m$, which are not necessarily distinct, are chosen randomly and independently from T, and for each k ($1 \leq k \leq m$), the set A_k is equally likely to be any of the sets in T.

(i) Write down the value of $\mathrm{P}(1 \in A_1)$.

(ii) By considering each integer separately, show that $\mathrm{P}(A_1 \cap A_2 = \varnothing) = \left(\frac{3}{4}\right)^n$. Find $\mathrm{P}(A_1 \cap A_2 \cap A_3 = \varnothing)$ and $\mathrm{P}(A_1 \cap A_2 \cap \cdots \cap A_m = \varnothing)$.

(iii) Find $\mathrm{P}(A_1 \subseteq A_2)$, $\mathrm{P}(A_1 \subseteq A_2 \subseteq A_3)$ and $\mathrm{P}(A_1 \subseteq A_2 \subseteq \cdots \subseteq A_m)$.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself How can you describe a subset of $\{1, \ldots, n\}$ by a sequence of n answers to yes/no questions?

STAGE 2 Disjoint sets

Ask yourself For $A_1 \cap A_2 = \varnothing$, what must be true of each element i? What is the probability of that, for one element? Why can you multiply over the elements?

STAGE 3 Subsets

Ask yourself For $A_1 \subseteq A_2$, which of the four possibilities for an element is forbidden?

Ask yourself For $A_1 \subseteq A_2 \subseteq A_3$, which of the 8 patterns of membership (in or out of each set) are allowed for an element?

Try this next

1 **Covering.** Find $\mathrm{P}(A_1 \cup A_2 = S)$ and $\mathrm{P}(A_1 \cup A_2 \cup \cdots \cup A_m = S)$.

2 **Same size.** Show that $\mathrm{P}(|A_1| = |A_2|) = \binom{2n}{n}/4^n$. (*Hint:* $|A_1| = |A_2|$ exactly when $|A_1| + |S \setminus A_2| = n$.) Show that this is about $1/\sqrt{\pi n}$ for large n.

3 **Partitions into three.** Show that the number of ordered pairs of disjoint subsets (A, B) of S is 3^n, and find the number of unordered pairs $\{A, B\}$ of disjoint subsets.

4 **Sperner for small n.** For $n = 3$ and $n = 4$, find the largest family of subsets of $\{1, \ldots, n\}$ in which no set contains another.

Further reading

- **Miklós Bóna, *A Walk Through Combinatorics*** (World Scientific), the chapters on subsets and partially ordered sets.
- **Béla Bollobás, *Combinatorics: Set Systems, Hypergraphs, Families of Vectors and Combinatorial Probability*** (Cambridge University Press). A short, elegant book on families of subsets, from Sperner's theorem onwards.

- **Peter Cameron, *Combinatorics: Topics, Techniques, Algorithms*** (Cambridge University Press). A first course with a strong emphasis on counting in two ways.

The full guided solution, and what the question leads to at university, is in Section 9.12:
https://beyondstep.keithmcnulty.org/ch09/random-subsets.html